

27-2 Port Security Troubleshooting

Answer Key

In this lab you will troubleshoot Port Security on a small campus network.

This lab continues on from the previous '28-1 Port Security Configuration'. Please ensure you have configured the Port Security settings in that lab.

Two additional PCs have been added, PC3 and PC4. The new PCs are not currently connected to the network.

In order for the lab to work as described please follow the steps in the exact order shown.

Port Security Troubleshooting

- 1) Verify that you can ping PC1 from PC2.

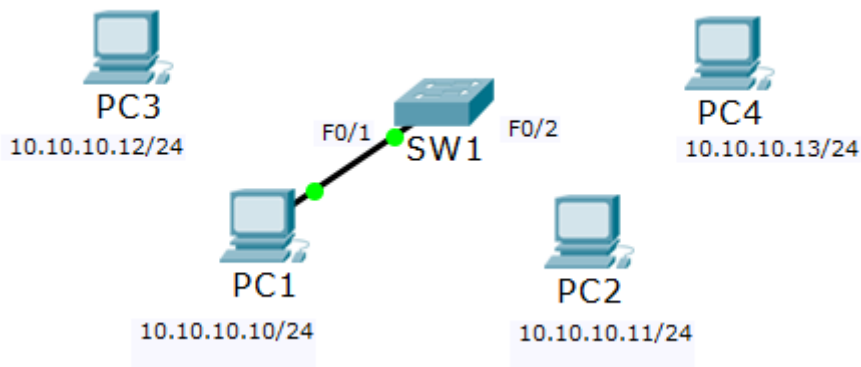
```
C:\>ping 10.10.10.10

Pinging 10.10.10.10 with 32 bytes of data:

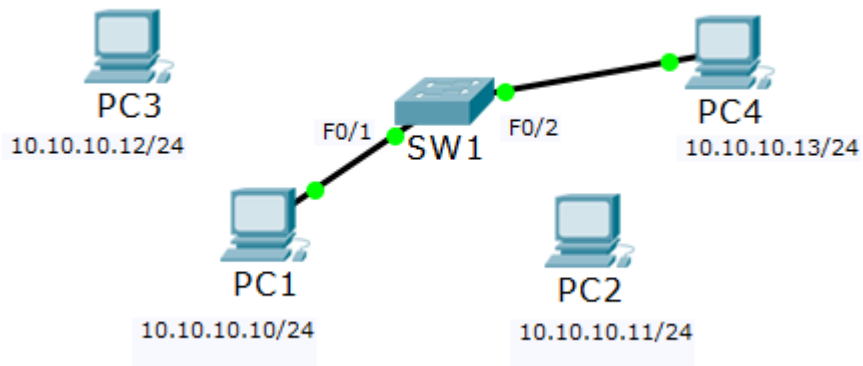
Reply from 10.10.10.10: bytes=32 time=13ms TTL=128
Reply from 10.10.10.10: bytes=32 time=3ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 4ms
```

- 2) In Packet Tracer, delete the Ethernet cable connecting SW1 interface FastEthernet 0/2 to PC2.



- 3) Connect interface FastEthernet 0/2 on SW1 to the FastEthernet0 port on PC4.



- 4) Do you expect a ping from the PC4 to PC1 at 10.10.10.10 to succeed?
Why or why not? Verify this.

The ping will succeed. SW1 will learn the new MAC address when PC4 is plugged in and generates traffic. SW1 receives traffic on the port from the allowed maximum of one MAC address. The traffic will be allowed.

```
SW1#show port-security int f0/2
Port Security : Enabled
Port Status : Secure-up
Violation Mode : Shutdown
Aging Time : 0 mins
Aging Type : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses : 1
Total MAC Addresses : 1
Configured MAC Addresses : 0
Sticky MAC Addresses : 0
Last Source Address:Vlan : 0000.4444.4444:1
Security Violation Count : 0
```

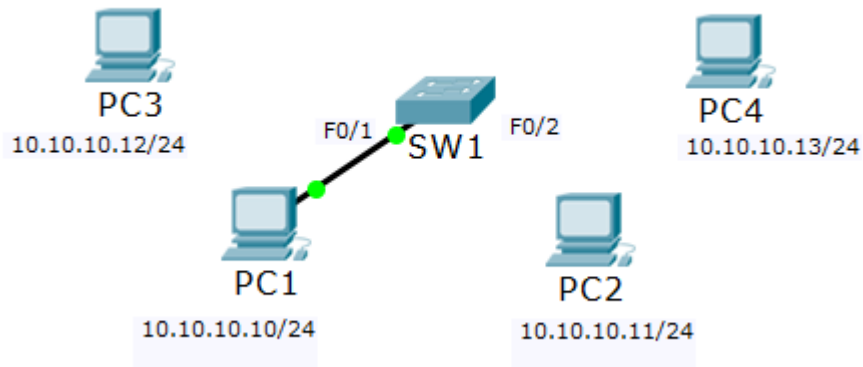
```
C:\>ping 10.10.10.10

Pinging 10.10.10.10 with 32 bytes of data:

Reply from 10.10.10.10: bytes=32 time=13ms TTL=128
Reply from 10.10.10.10: bytes=32 time=3ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 13ms, Average = 4ms
```

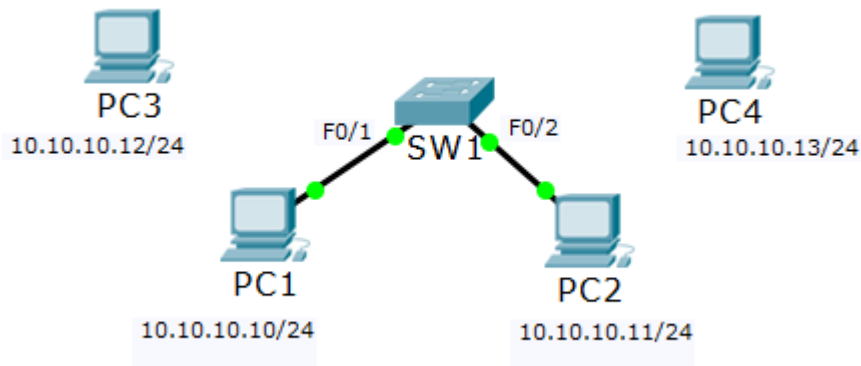
- 5) In Packet Tracer, delete the Ethernet cable connecting SW1 interface FastEthernet 0/2 to PC4.



- 6) Enable Port Security sticky MAC address learning on SW1 interface FastEthernet 0/2.

```
SW1(config)#interface f0/2
SW1(config-if)#switchport port-security mac-address sticky
```

- 7) Connect interface FastEthernet 0/2 on SW1 to the FastEthernet0 port on PC2.



- 8) Do you expect a ping from the PC2 to PC1 at 10.10.10.10 to succeed? Why or why not? Verify this.

The ping will succeed. SW1 will learn the PC2's MAC address when it is plugged in and generates traffic. SW1 receives traffic on the port from the allowed maximum of one MAC address. The traffic will be allowed.

PC2's MAC address is now added as a sticky MAC address.

```
SW1#sh port-security int f0/2
Port Security : Enabled
Port Status : Secure-up
Violation Mode : Shutdown
Aging Time : 0 mins
Aging Type : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses : 1
Total MAC Addresses : 1
Configured MAC Addresses : 0
Sticky MAC Addresses : 1
Last Source Address:Vlan : 0000.2222.2222:1
Security Violation Count : 0
```

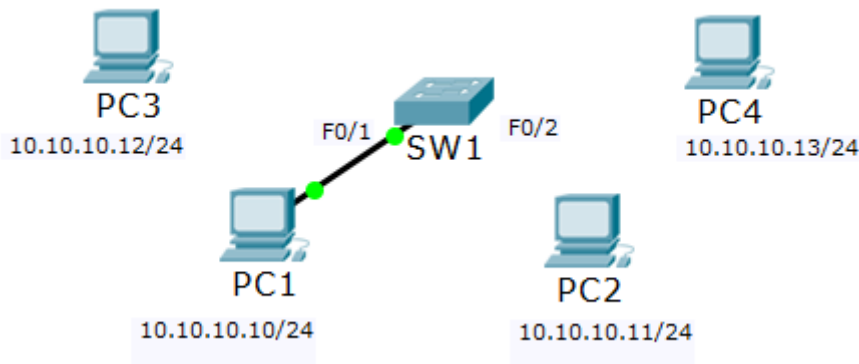
```
C:\>ping 10.10.10.10

Pinging 10.10.10.10 with 32 bytes of data:

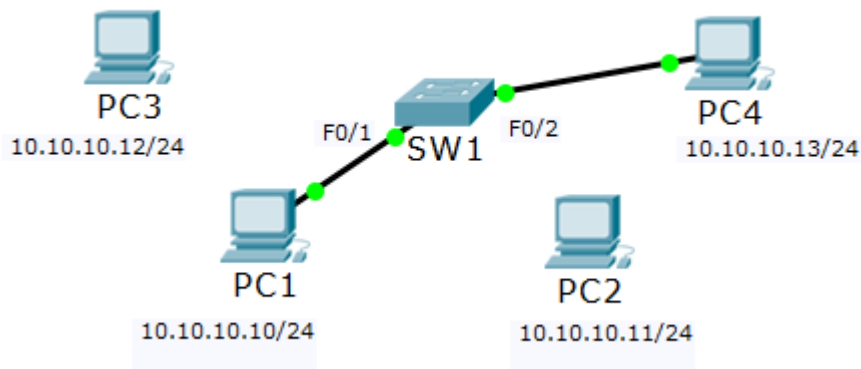
Reply from 10.10.10.10: bytes=32 time=13ms TTL=128
Reply from 10.10.10.10: bytes=32 time=3ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 4ms
```

- 9) In Packet Tracer, delete the Ethernet cable connecting SW1 interface FastEthernet 0/2 to PC2.



10) Connect interface FastEthernet 0/2 on SW1 to the FastEthernet0 port on PC4.



11) Do you expect a ping from the PC4 to PC1 at 10.10.10.10 to succeed? Why or why not? Verify this.

The ping will fail. SW1 already added PC2's MAC address as a sticky MAC address and it is taking up the maximum one allowed MAC address.

```
C:\>ping 10.10.10.10

Pinging 10.10.10.10 with 32 bytes of data:

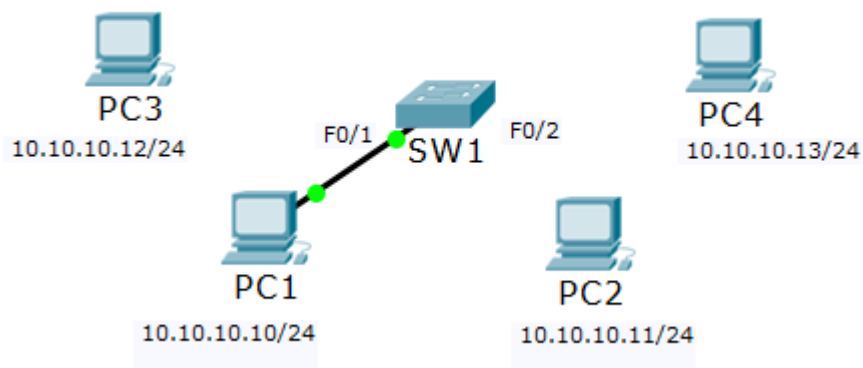
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

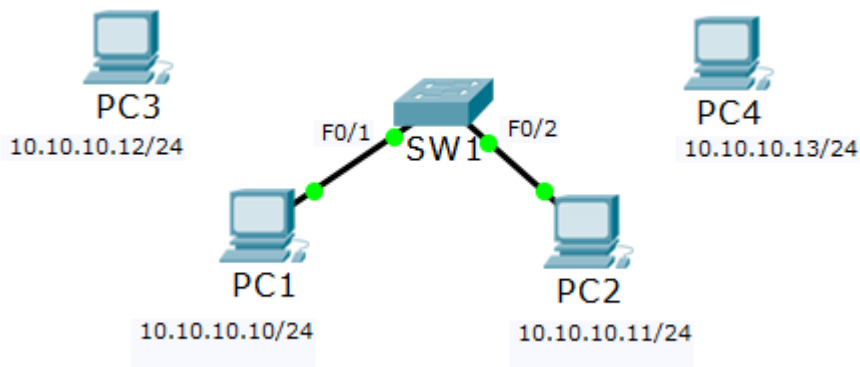
12) What is the Port Security status on SW1 interface FastEthernet 0/2?

```
SW1#sh port-security int f0/2
Port Security : Enabled
Port Status : Secure-shutdown
Violation Mode : Shutdown
Aging Time : 0 mins
Aging Type : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses : 1
Total MAC Addresses : 1
Configured MAC Addresses : 0
Sticky MAC Addresses : 1
Last Source Address:Vlan : 0000.4444.4444:1
Security Violation Count : 1
```

13) In Packet Tracer, delete the Ethernet cable connecting SW1 interface FastEthernet 0/2 to PC4.



14) Connect interface FastEthernet 0/2 on SW1 to the FastEthernet0 port on PC2.



- 15) Do you expect a ping from the PC2 to PC1 at 10.10.10.10 to succeed?
Why or why not? Verify this.

The ping will fail. Even though the interface is connected back to the PC with the correct MAC address, the interface will remain error-disabled. You need to manually shutdown then no shutdown the port to bring it back up.

```
SW1#sh port-security int f0/2
Port Security : Enabled
Port Status : Secure-shutdown
Violation Mode : Shutdown
Aging Time : 0 mins
Aging Type : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses : 1
Total MAC Addresses : 1
Configured MAC Addresses : 0
Sticky MAC Addresses : 1
Last Source Address:Vlan : 0000.4444.4444:1
Security Violation Count : 1
```

```
C:\>ping 10.10.10.10

Pinging 10.10.10.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- 16) Ensure that PC2 can ping PC1. Do not change any 'switchport port-security' commands to do this.

```
SW1(config)#int f0/2
SW1(config-if)#shutdown
SW1(config-if)#no shutdown
```



```
SW1#sh port-security int f0/2
Port Security : Enabled
Port Status : Secure-up
Violation Mode : Shutdown
Aging Time : 0 mins
Aging Type : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses : 1
Total MAC Addresses : 1
Configured MAC Addresses : 0
Sticky MAC Addresses : 1
Last Source Address:Vlan : 0000.2222.2222:1
Security Violation Count : 0
```

```
C:\>ping 10.10.10.10

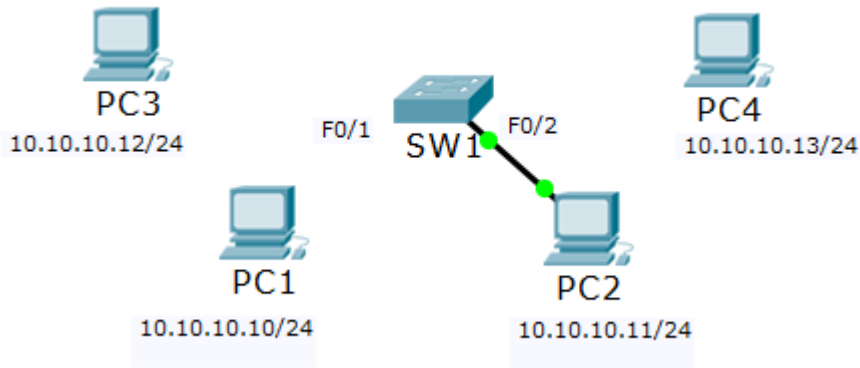
Pinging 10.10.10.10 with 32 bytes of data:

Reply from 10.10.10.10: bytes=32 time=13ms TTL=128
Reply from 10.10.10.10: bytes=32 time=3ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128
Reply from 10.10.10.10: bytes=32 time<1ms TTL=128

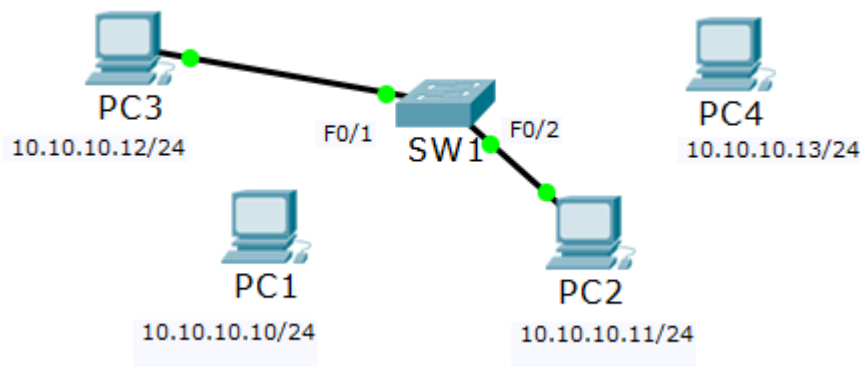
Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 13ms, Average = 4ms
```

- 17) **PC1's MAC address 0000.1111.1111 has been manually added to the Port Security configuration on SW1 interface FastEthernet 0/1.**

In Packet Tracer, delete the Ethernet cable connecting SW1 interface FastEthernet 0/1 to PC1.



- 18) Connect interface FastEthernet 0/1 on SW1 to the FastEthernet0 port on PC3.



- 19) Do you expect a ping from the PC3 to PC2 at 10.10.10.11 to succeed? Why or why not? Verify this.

The ping will succeed. Interface F0/1 has been configured to allow a maximum of two MAC addresses. PC1's manually added address is taking up one of the allocation. When the cable is moved to PC3 the switch will learn the new MAC address. The maximum allocation has now been used but not exceeded.

```
SW1#sh port-security int f0/1
Port Security : Enabled
Port Status : Secure-up
Violation Mode : Shutdown
Aging Time : 0 mins
Aging Type : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses : 2
Total MAC Addresses : 2
Configured MAC Addresses : 1
Sticky MAC Addresses : 0
Last Source Address:Vlan : 0000.3333.3333:1
Security Violation Count : 0
```

```
C:\>ping 10.10.10.11

Pinging 10.10.10.11 with 32 bytes of data:

Reply from 10.10.10.11: bytes=32 time=11ms TTL=128
Reply from 10.10.10.11: bytes=32 time=21ms TTL=128
Reply from 10.10.10.11: bytes=32 time<1ms TTL=128
Reply from 10.10.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 21ms, Average = 8ms
```